

Quiz #3

C8-D

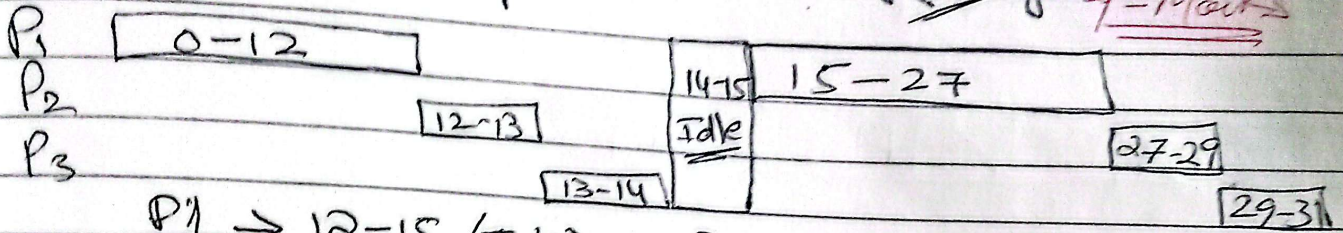
Operating Systems

Solution

Arrival time	δ	P1	12	3	12	7.5 $\rightarrow$ FCFS
1		P2	1	2	2	7.5 $\rightarrow$ RR
2		P3	1	2	2	15 $\rightarrow$ Total Marks
		CPU	I/O	CPU		

FCFS (non preemptive):

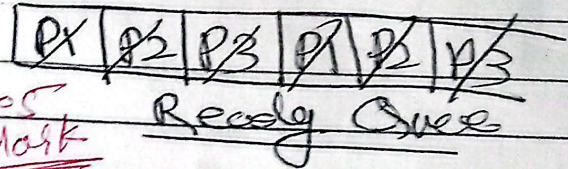
Running 4-Marks



$P_1 \rightarrow 12-15$ (I/O)

$P_2 \rightarrow 13-15$ (I/O)

$P_3 \rightarrow 14-16$ (I/O)



P1

0.5 Mark

Ready Queue

Waiting Time $\Rightarrow \frac{0}{0-12} + \frac{0}{15-27} = 0$

Turnaround Time $\Rightarrow 27-0 \Rightarrow 27$

P2

1 Mark

Waiting Time $\Rightarrow \frac{11}{1-12} + \frac{12}{15-27} = 23$

Turnaround Time $\Rightarrow 29-1 \Rightarrow 28$

P3

1 Mark

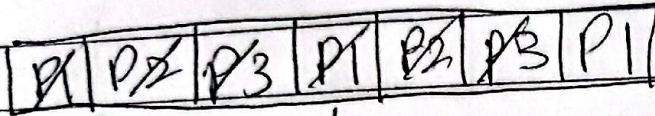
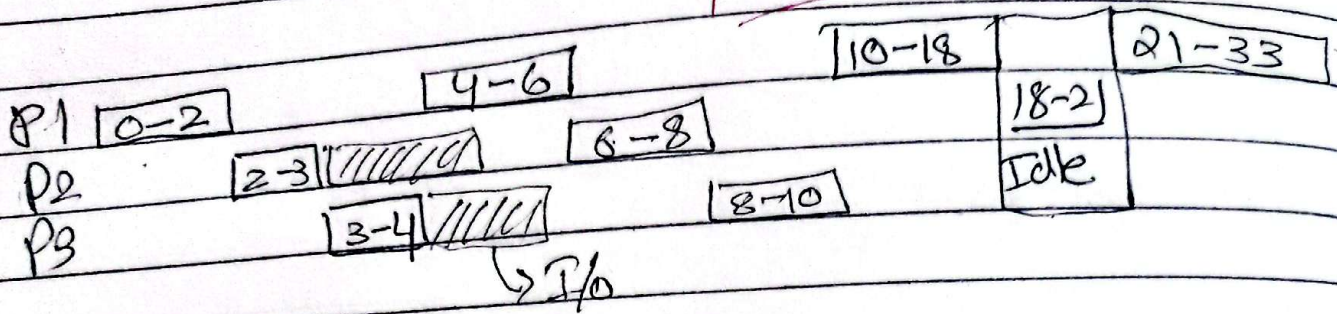
Waiting Time $\Rightarrow \frac{11}{2-13} + \frac{13}{16-29} = 24$

Turnaround Time $\Rightarrow 31-2 = 29$

Avg. waiting = $\frac{0+23+24}{3} \Rightarrow 15.67$ Avg. TAT $\Rightarrow \frac{27+28+29}{3} = 28$

Round Robin (Q=2)

4-Marks



0.5 Marks

↓ both P2/P3 executed

P1 → 18-21 (I/O) P2 → 3-5 (I/O)
P3 → 4-6 (I/O)

P1

1 Mark waiting time ⇒ $\frac{2}{2-4} + \frac{4}{6-10} = 6$

TAT ⇒ $33-0 \Rightarrow 33$

P2

1 Mark waiting time ⇒ $\frac{1}{1-2} + \frac{1}{5-6} \Rightarrow 2$

TAT ⇒ $8-1 \Rightarrow 7$

P3

1 Mark waiting time ⇒ $\frac{1}{2-3} + \frac{2}{6-8} = 3$

TAT ⇒ $10-2 \Rightarrow 8$

Avg waiting time ⇒ $6+2+3 \Rightarrow 3.67$

Avg TAT ⇒ $\frac{33+7+8}{3} \Rightarrow 16$